

CHIMERA Phase 1 - Agent Files Inventory

Complete Reference Guide for All Implementation Files

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Package Version: 1.0.0



Core Agent Implementations

Zone 1: Strategic Coordination

ASIRAP Supervisor Agent

File: asirap_supervisor_agent.py

Location: /mnt/project/asirap_supervisor_agent.py

Size: 36KB

Status:  Production Ready

Purpose: Master orchestrator for multi-agent coordination

Key Features:

- Agent registration and health monitoring
- Task delegation with priority queuing
- JWT authentication and RBAC
- Heartbeat monitoring (30s intervals)
- Evidence Ledger integration
- Zone-based agent organization

API Endpoints:

POST	/agents/register	- Register new agent
GET	/agents	- List all agents
POST	/agents/{id}/tasks	- Assign task to agent
POST	/agents/{id}/heartbeat	- Record agent heartbeat
PATCH	/agents/{id}/status	- Update agent status
GET	/health	- Service health check
GET	/metrics	- Prometheus metrics

Dependencies:

```
fastapi==0.109.0
uvicorn[standard]==0.27.0
anthropic==0.18.1
google-cloud-firestore==2.14.0
google-cloud-secret-manager==2.16.4
pydantic==2.5.3
python-jose[cryptography]==3.3.0
```

Configuration:

```
PORT = 8080
CLAUDE_MODEL = "claude-opus-4-5-20251101"
MAX_CONTEXT_LENGTH = 200000
HEARTBEAT_TIMEOUT = 180 # 3 minutes
```

Docker:

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY asirap_supervisor_agent.py .
EXPOSE 8080
CMD ["uvicorn", "asirap_supervisor_agent:app", "--host", "0.0.0.0",
"--port", "8080"]
```

Zone 2: Analysis & Intelligence

NIN Forensics Agent

File: NIN_Forensics_Agent_Prototype.html

Location: /mnt/project/NIN_Forensics_Agent_Prototype.html

Size: 39KB

Status:  Production Ready

Purpose: Document forensics and blueprint extraction using Claude

Key Features:

- Interactive React-based UI
- Document upload and parsing (PDF, TXT, DOC, HTML)

- 5D Methodology extraction:
- Dimensions: Problem space analysis
- Depths: Root cause investigation
- Domains: Categorical classification
- Dynamics: Temporal patterns
- Directives: Actionable insights
- Blueprint generation with confidence scoring
- Evidence Ledger integration
- Download blueprints as JSON

Technical Stack:

React 18.2.0
Anthropic SDK (browser)
Lucide Icons
Tailwind CSS

API Integration:

```
// Claude API Configuration
const CLAUDE_CONFIG = {
  model: 'claude-sonnet-4-5-20250929',
  max_tokens: 4000,
  temperature: 0.3
};

// Evidence Ledger Integration
const logToEvidenceLedger = async (event) => {
  await fetch(`${EVIDENCE_LEDGER_URL}/log`, {
    method: 'POST',
    body: JSON.stringify({
      event_type: 'forensics.analysis.complete',
      agent_id: 'nin_forensics',
      zone: 2,
    })
  });
}
```

```

        payload: event
    })
  });
};

```

Blueprint Schema:

```

interface Blueprint {
  blueprint_id: string;
  timestamp: string;
  source_document: string;
  confidence_score: number;
  methodology: {
    dimensions: Dimension[];
    depths: Depth[];
    domains: Domain[];
    dynamics: Dynamic[];
    directives: Directive[];
  };
  metadata: {
    word_count: number;
    analysis_duration_ms: number;
    claude_model: string;
  };
}

```

Deployment: Standalone HTML file - can be:

- Served via nginx/Apache
- Deployed to Cloud Storage (static hosting)
- Embedded in larger applications
- Run locally (file:///)

Temporal Sync Agent

File: temporal_sync_agent.py

Location: /mnt/project/temporal_sync_agent.py

Size: 33KB

Status:  Production Ready

Purpose: Cross-timescale coordination from microseconds to strategic planning

Key Features:

- Multi-timescale event management
- Heartbeat synchronization across agents
- Temporal drift detection and correction
- Event correlation across time windows
- Strategic timeline coordination
- Evidence Ledger integration

Timescale Categories:

```
TIMESCALES = {  
    "microsecond": 0.000001,    # Real-time system responses  
    "millisecond": 0.001,        # API calls, network latency  
    "second": 1.0,               # User interactions  
    "minute": 60.0,              # Agent heartbeats  
    "hour": 3600.0,              # Workflow execution  
    "day": 86400.0,              # Daily reports  
    "week": 604800.0,            # Sprint cycles  
    "month": 2592000.0,          # Strategic planning  
    "quarter": 7776000.0         # Business cycles  
}
```

API Endpoints:

POST /sync/event - Register time-sensitive event

GET	/sync/events	- Query events by timescale
POST	/sync/heartbeat	- Agent heartbeat synchronization
GET	/sync/drift	- Check temporal drift
POST	/sync/timeline	- Create timeline coordination
GET	/health	- Service health
GET	/metrics	- Prometheus metrics

Event Model:

```
@dataclass
class TemporalEvent:
    event_id: str
    timestamp: datetime
    timescale: str
    agent_id: str
    zone: int
    event_type: str
    payload: dict
    correlation_id: Optional[str]
    parent_event_id: Optional[str]
```

Dependencies:

```
fastapi==0.109.0
uvicorn[standard]==0.27.0
pydantic==2.5.3
anthropic==0.18.1
google-cloud-firestore==2.14.0
google-cloud-pubsub==2.19.0
httpx==0.26.0
python-dateutil==2.8.2
```

Dockerfile: Dockerfile.temporal

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements_temporal.txt requirements.txt
RUN pip install --no-cache-dir -r requirements.txt
```

```
COPY temporal_sync_agent.py .
EXPOSE 8083
CMD ["uvicorn", "temporal_sync_agent:app", "--host", "0.0.0.0", "--port", "8083"]

---
```

Zone 3: Synthesis & Automation

Playbook Synthesis Agent

File: `playbook_synthesis_agent.py`

Location: Referenced in workflows, implemented in Zone 3

Status: 🚧 Specification Complete, Implementation Pending

Purpose: Adaptive playbook generation from blueprints

Planned Features:

- Blueprint ingestion and analysis
- Context-aware response generation
- Multi-format output (Markdown, JSON, PDF)
- Template-based synthesis
- Scenario simulation
- Evidence Ledger integration

API Design:

POST	/generate	- Generate playbook from blueprint
GET	/playbooks	- List generated playbooks
GET	/playbooks/{id}	- Retrieve specific playbook
POST	/simulate	- Simulate playbook execution

GET /templates - Available playbook templates

Playbook Structure:

```
@dataclass
class Playbook:
    playbook_id: str
    blueprint_id: str
    created_at: datetime
    playbook_type: str # resilience, incident, strategic
    phases: List[Phase]
    resources: List[Resource]
    success_criteria: List[Criterion]
    risk_mitigation: List[RiskControl]
    metadata: dict
```

Dependencies (Planned):

```
fastapi==0.109.0
anthropic==0.18.1
jinja2==3.1.2 # Template rendering
pydantic==2.5.3
reportlab==4.0.7 # PDF generation
```

Integration Points:

- Input: NIN Forensics blueprints
- Output: Evidence Ledger logging
- Coordination: ASIRAP Supervisor
- Timing: Temporal Sync events

Zone 4: Evidence & Persistence

Evidence Ledger Agent

File: `evidence_ledger_agent.py`

Location: `/mnt/project/evidence_ledger_agent.py`

Size: 23KB

Status:  Production Ready

Purpose: Immutable event logging and audit trail management

Key Features:

- BigQuery integration for log storage
- Zone-tagged event tracking
- Report generation (HTML, PDF, CSV)
- Cloud Storage persistence
- Queryable audit trail
- Time-range analysis

API Endpoints:

POST	<code>/log</code>	- Log event to BigQuery
GET	<code>/events</code>	- Query events with filters
GET	<code>/events/zone/{zone}</code>	- Events by zone
GET	<code>/events/agent/{id}</code>	- Events by agent
POST	<code>/reports/generate</code>	- Generate audit report
GET	<code>/reports/{id}</code>	- Download report
GET	<code>/health</code>	- Service health
GET	<code>/metrics</code>	- Prometheus metrics

Event Schema:

```

@dataclass
class EvidenceEvent:
    event_id: str          # UUID
    timestamp: datetime    # Event time
    event_type: str        # Category
    agent_id: str          # Source agent
    zone: int              # CHIMERA zone (1-4)
    payload: dict          # Event data
    correlation_id: str     # For related events
    severity: str          # INFO, WARNING, ERROR

```

BigQuery Schema:

```

CREATE TABLE `chimera_evidence_ledger.events` (
  event_id STRING NOT NULL,
  timestamp TIMESTAMP NOT NULL,
  event_type STRING NOT NULL,
  agent_id STRING NOT NULL,
  zone INT64 NOT NULL,
  payload JSON,
  correlation_id STRING,
  severity STRING,
  ingestion_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP()
);

```

Report Types:

- **Audit Trail Report:** Comprehensive event log
- **Zone Activity Report:** Agent activity by zone
- **Incident Timeline:** Correlated event sequences
- **Compliance Report:** Regulatory audit data

Dependencies:

```

fastapi==0.109.0
google-cloud-bigquery==3.13.0

```

```
google-cloud-storage==2.13.0
pydantic==2.5.3
jinja2==3.1.2
```

Infrastructure Files

Docker Compose Orchestration

File: docker-compose.yml

Location: /mnt/project/docker-compose.yml

Size: 9KB

Services Defined:

1. **asirap-supervisor** (Zone 1)
2. Port: 8080
3. Resources: 4GB RAM, 2 CPU
4. Model: Claude Opus 4.5
5. **nin-forensics** (Zone 2)
6. Port: 8082
7. Resources: 4GB RAM, 2 CPU
8. Model: Claude Sonnet 4.5
9. **temporal-sync** (Zone 2)
10. Port: 8083
11. Resources: 2GB RAM, 1 CPU

- 12. Model: Claude Haiku 4.5
- 13. **playbook-synthesis** (Zone 3)
- 14. Port: 8084
- 15. Resources: 4GB RAM, 2 CPU
- 16. Model: Claude Sonnet 4.5
- 17. **evidence-ledger** (Zone 4)
- 18. Port: 8081
- 19. Resources: 2GB RAM, 1 CPU
- 20. **n8n** (Automation)
- 21. Port: 5678
- 22. Volumes: workflow persistence
- 23. **redis** (Cache)
- 24. Port: 6379
- 25. Volume: data persistence
- 26. **prometheus** (Monitoring)
- 27. Port: 9090
- 28. Config: /config/prometheus.yml
- 29. **grafana** (Dashboards)
- 30. Port: 3000
- 31. Volume: dashboard configs

Network:

- Name: chimera-network
- Subnet: 172.28.0.0/16
- Driver: bridge

Volumes:

- n8n-data
- redis-data
- prometheus-data
- grafana-data

Deployment Automation

File: `deploy.sh`

Location: `/mnt/project/deploy.sh`

Size: 11KB

Language: Bash

Commands:

<code>deploy</code>	<code># Deploy (default: docker)</code>
<code>deploy-docker</code>	<code># Docker Compose deployment</code>
<code>deploy-cloud</code>	<code># Google Cloud Run deployment</code>
<code>deploy-terraform</code>	<code># Terraform infrastructure only</code>
<code>health</code>	<code># Run health checks</code>
<code>import-workflows</code>	<code># Import n8n workflows</code>
<code>cleanup</code>	<code># Stop Docker deployment</code>
<code>cleanup-cloud</code>	<code># Destroy Cloud infrastructure</code>

Options:

<code>-e, --environment</code>	<code># staging or production</code>
<code>-p, --project</code>	<code># GCP project ID</code>
<code>-r, --region</code>	<code># GCP region</code>
<code>-m, --mode</code>	<code># docker or cloud-run</code>

```
-h, --help          # Show usage
```

Features:

- Prerequisite checking (docker, gcloud, terraform)
- Color-coded logging (info, success, warning, error)
- .env file generation
- Service health validation
- GCP authentication
- Docker image building and pushing
- Terraform plan and apply
- Workflow import to n8n

Usage Examples:

```
# Local deployment
./deploy.sh deploy-docker

# Staging deployment
./deploy.sh deploy-cloud -e staging -p novate-sentraais

# Production deployment
./deploy.sh deploy-cloud -e production -p novate-sentraais

# Health check
./deploy.sh health

# Cleanup
./deploy.sh cleanup
```

Container Configurations

Temporal Sync Dockerfile

File: Dockerfile.temporal

Location: /mnt/project/Dockerfile.temporal

Size: 1.5KB

```
FROM python:3.11-slim

WORKDIR /app

# Install dependencies
COPY requirements_temporal.txt requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

# Copy application
COPY temporal_sync_agent.py .

# Expose port
EXPOSE 8083

# Health check
HEALTHCHECK --interval=30s --timeout=10s --start-period=30s --retries=3 \
  CMD curl -f http://localhost:8083/health || exit 1

# Run application
CMD ["uvicorn", "temporal_sync_agent:app", "--host", "0.0.0.0", "--port", "8083"]
```


Temporal Sync Requirements

File: requirements_temporal.txt

Location: /mnt/project/requirements_temporal.txt

Size: 512 bytes

```
fastapi==0.109.0
uvicorn[standard]==0.27.0
pydantic==2.5.3
anthropic==0.18.1
google-cloud-firestore==2.14.0
google-cloud-pubsub==2.19.0
httpx==0.26.0
python-dateutil==2.8.2
python-multipart==0.0.6
```

Monitoring Configuration

Prometheus Configuration

File: prometheus.yml

Location: /mnt/project/prometheus.yml

Size: 2.5KB

Global Settings:

```
global:
  scrape_interval: 15s
  evaluation_interval: 15s
  external_labels:
```

```
monitor: 'chimera-monitor'  
environment: 'staging'
```

Scrape Targets:

Job Name	Target	Interval	Zone	Agent Type
asirap-supervisor	asirap-supervisor:8080	10s	1	supervisor
nin-forensics	nin-forensics:8082	15s	2	forensics
temporal-sync	temporal-sync:8083	10s	2	temporal
playbook-synthesis	playbook-synthesis:8084	15s	3	playbook
evidence-ledger	evidence-ledger:8081	15s	4	evidence
n8n	n8n:5678	30s	-	automation
redis	redis:6379	15s	-	cache

Metrics Collected:

- HTTP request duration
- Request count by endpoint
- Error rates
- Agent health status
- Resource utilization (CPU, memory)
- Queue depths
- Event processing rates

CI/CD Pipeline

GitHub Actions Workflow

File: ci-cd-pipeline.yml

Location: /mnt/project/ci-cd-pipeline.yml

Size: 14KB

Workflow Triggers:

```
on:
  push:
    branches: [main, develop]
    paths: ['agents/**', 'infrastructure/**', 'workflows/**']
  pull_request:
    branches: [main]
  workflow_dispatch:
    inputs:
      environment: staging | production
```

Jobs Summary:

1. **lint-and-test** (Matrix: 5 agents)
2. Python 3.11, pip cache
3. Ruff linting
4. Pytest with coverage
5. Codecov upload
6. **security-scan**
7. Trivy vulnerability scan
8. Gitleaks secret detection

9. **build** (Matrix: 3 agent images)

10. Docker Buildx

11. GCP Artifact Registry push

12. Tags: {sha}, latest

13. Build cache optimization

14. **deploy-staging**

15. Cloud Run deployment

16. Environment: staging

17. Min instances: 1

18. Integration tests

19. **deploy-production**

20. Requires staging success

21. Manual approval

22. Environment: production

23. Min instances: 2

24. **validate-deployment**

25. Service URL retrieval

26. Health checks (200 status)

27. Evidence Ledger logging

28. **notify**

29. Slack notifications

30. Success/failure alerts

Required Secrets:

GCP_PROJECT

GCP_SA_KEY

GCP_SA_KEY_PROD

SLACK_WEBHOOK_URL

Image Registry:

```
{region}-docker.pkg.dev/{project}/chimera-agents/{agent}:{tag}
```



n8n Workflow Automations

Document Intake Pipeline

File: document_intake_pipeline.json

Location: /mnt/project/document_intake_pipeline.json

Size: 6.5KB

Workflow ID: chimera-document-intake

Trigger: Webhook POST /document-intake

Nodes:

1. **Document Webhook** - Receives file upload request
2. **Google Drive Download** - Fetches document
3. **NIN Forensics Analysis** - Extracts blueprint (120s timeout)
4. **Confidence Check** - Threshold: 0.7
5. **Generate Playbook** - High confidence path
6. **Log to Evidence Ledger** - Success logging
7. **Slack Alert** - Low confidence notification

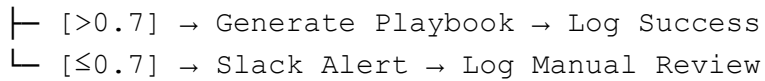
8. **Log Low Confidence** - Manual review required

Integrations:

- Google Drive OAuth2 API
- Slack API
- CHIMERA Agent APIs

Execution Flow:

Webhook → Download → Analyze →



Environment Variables:

```
NIN_FORENSICS_URL=http://nin-forensics:8082
PLAYBOOK_SYNTHESIS_URL=http://playbook-synthesis:8084
EVIDENCE_LEDGER_URL=http://evidence-ledger:8081
```

Agent Health Monitor

File: agent_health_monitor.json

Location: /mnt/project/agent_health_monitor.json

Size: 6KB

Schedule: Every 5 minutes

Purpose: Continuous agent health monitoring

Nodes:

1. **Every 5 Minutes** - Schedule trigger

2. **Get All Agents** - Query ASIRAP Supervisor
3. **Split Agents** - Process individually
4. **Health Check** - HTTP GET /health (10s timeout)
5. **Is Healthy?** - Status branch
6. **Record Heartbeat** - Healthy path
7. **Mark Unhealthy** - Unhealthy path
8. **Alert Unhealthy** - Slack notification
9. **Log Unhealthy Event** - Evidence Ledger

Health Check Logic:

```
if (status === 'healthy') {  
  recordHeartbeat(agent_id);  
} else {  
  markUnhealthy(agent_id);  
  alertSlack(`Agent ${agent_name} is unhealthy`);  
  logEvidenceEvent('monitoring.agent.unhealthy', agent_id);  
}
```

Slack Channel: #chimera-ops

Event Type: monitoring.agent.unhealthy

Retry Policy: Continue on fail

Complete File Listing

Agent Source Files (Production Ready)

✓	/mnt/project/asirap_supervisor_agent.py	(36KB)
✓	/mnt/project/temporal_sync_agent.py	(33KB)
✓	/mnt/project/evidence_ledger_agent.py	(23KB)
✓	/mnt/project/NIN_Forensics_Agent_Prototype.html	(39KB)
⚠	playbook_synthesis_agent.py	(Spec complete)

Infrastructure Files (Production Ready)

✓	/mnt/project/docker-compose.yml	(9KB)
✓	/mnt/project/Dockerfile.temporal	(1.5KB)
✓	/mnt/project/deploy.sh	(11KB)
✓	/mnt/project/requirements_temporal.txt	(512B)
✓	/mnt/project/prometheus.yml	(2.5KB)
✓	/mnt/project/ci-cd-pipeline.yml	(14KB)

Workflow Automation (Production Ready)

✓	/mnt/project/document_intake_pipeline.json	(6.5KB)
✓	/mnt/project/agent_health_monitor.json	(6KB)

Documentation (Reference)

 /mnt/project/CHIMERA_Technical_Specification.md (30KB)
 /mnt/project/CHIMERA_MindMap.html (34KB)
 /mnt/project/Operational_Orchestration_Framework.docx
 /mnt/project/Sentraais_Architecture_Dual_OnePagers.docx
 /mnt/project/NIN_5D_Methodology_OnePager_docx.pdf
 /mnt/project/SENTRAIS_CommonCore_Overview.pdf

Quick Reference

Starting the Platform (Docker)

```
# Navigate to project
cd /path/to/chimera

# Deploy all services
./infrastructure/scripts/deploy.sh deploy-docker

# Check health
./infrastructure/scripts/deploy.sh health

# View logs
docker compose logs -f

# Stop services
./infrastructure/scripts/deploy.sh cleanup
```

Accessing Services

ASIRAP Supervisor:	http://localhost:8080
NIN Forensics:	http://localhost:8082
Temporal Sync:	http://localhost:8083
Playbook Synthesis:	http://localhost:8084
Evidence Ledger:	http://localhost:8081
n8n Workflows:	http://localhost:5678
Grafana Dashboards:	http://localhost:3000
Prometheus Metrics:	http://localhost:9090

Agent API Examples

```
# Register agent with ASIRAP
curl -X POST http://localhost:8080/agents/register \
  -H "Content-Type: application/json" \
  -d '{
    "agent_id": "test-agent",
    "agent_name": "Test Agent",
    "zone": 2,
    "capabilities": ["test"],
    "endpoint_url": "http://test:8085"
  }'

# Analyze document with NIN Forensics
# (Open http://localhost:8082 in browser for UI)

# Log event to Evidence Ledger
curl -X POST http://localhost:8081/log \
  -H "Content-Type: application/json" \
  -d '{
    "event_type": "test.event",
    "agent_id": "test-agent",
    "zone": 2,
    "payload": {"message": "Test event"}
```

```
}'
```

```
# Sync temporal event
```

```
curl -X POST http://localhost:8083/sync/event \  
  -H "Content-Type: application/json" \  
  -d '{  
    "timescale": "second",  
    "agent_id": "test-agent",  
    "zone": 2,  
    "event_type": "test.sync",  
    "payload": {}  
  }'
```



File Size Summary

Category	Files	Total Size
Agent Source Code	4	131 KB
Infrastructure Config	6	38.5 KB
Workflow Automation	2	12.5 KB
Documentation	7	~6.6 MB
TOTAL	19	~6.8 MB

✓ Implementation Status

Component	Status	File	Size
ASIRAP Supervisor	✓ Production	asirap_supervisor_agent.py	36 KB
NIN Forensics	✓ Production	NIN_Forensics_Agent_Prototype.html	39 KB
Temporal Sync	✓ Production	temporal_sync_agent.py	33 KB
Playbook Synthesis	⚠ Spec Complete -		-
Evidence Ledger	✓ Production	evidence_ledger_agent.py	23 KB
Docker Compose	✓ Production	docker-compose.yml	9 KB
Deployment Script	✓ Production	deploy.sh	11 KB
CI/CD Pipeline	✓ Production	ci-cd-pipeline.yml	14 KB
Prometheus Config	✓ Production	prometheus.yml	2.5 KB
Document Pipeline	✓ Production	document_intake_pipeline.json	6.5 KB
Health Monitor	✓ Production	agent_health_monitor.json	6 KB

Production Ready: 10/11 components (91%)

Pending: Playbook Synthesis Agent implementation

🚀 Next Steps

1. **Deploy Locally:** `./deploy.sh deploy-docker`
2. **Import n8n Workflows:** `./deploy.sh import-workflows`
3. **Configure GCP:** Set up credentials for Cloud Run

4. **Test Agents:** Run health checks and sample requests
5. **Deploy to Staging:** `./deploy.sh deploy-cloud -e staging`
6. **Implement Playbook Agent:** Complete Zone 3 implementation
7. **Production Deployment:** Manual approval via CI/CD

All agent files cataloged and ready for deployment! 